



Artificial Intelligence-based Rice Variety Classification: A State-of-the-art Review and Future Directions

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ABSTRACT

Rice is a staple food for a significant portion of the global population, making accurate classification of rice varieties essential for farming and consumer protection. This review provides a focused analysis of the current advancements and challenges in applying computer vision (CV) techniques to rice variety classification. The study examines key steps in the automation process, including image acquisition, pre-processing, feature extraction, and classification algorithms, with particular emphasis on machine learning and deep learning methods such as Convolutional Neural Networks (CNNs), which have demonstrated exceptional performance in recent research. However, practical implementation faces challenges, including the availability of high-quality datasets, the impact of environmental variations on image quality, and the computational demands of complex models. Our study discusses these obstacles and highlights the importance of developing resilient and scalable systems for real-world applications. By synthesizing findings from various studies, this review proposes future directions for advancing rice variety classification, focusing on improved feature extraction techniques, enhanced dataset management, and integrating innovative machine learning paradigms. This work is a valuable resource for researchers and practitioners aiming to advance rice classification technologies and contribute to food security and agricultural sustainability.

1. Introduction

Rice (*Oryza sativa*) is one of the most consumed staple foods in the world, sustaining more than half of the global population. In Asia, rice is the primary source of calories and nutrients, making it a cornerstone of nutrition, food security, economic development, and livelihoods for millions of farmers [1]. With approximately 40,000 rice varieties under cultivation globally [2], the accurate identification and classification of these varieties are critical for ensuring quality, preventing fraud, and advancing agricultural research [3]. The classification of rice varieties presents significant challenges due to the subtle morphological variations between varieties that are often indistinguishable from the human eye [4]. Furthermore, environmental factors—such as soil type, climatic conditions, and farming practices—can influence the physical traits of rice grains, further complicating the task. Traditional methods of rice identification, which rely on expert visual inspection, are

time-consuming, labor-intensive, and prone to errors [5]. As the global demand for high-quality food grows, these limitations highlight the pressing need for effective, robust, and scalable alternatives. Recent advancements in computer vision technologies have offered promising solutions for automating the classification of rice varieties. These systems utilize visual data and machine learning algorithms to analyze images of rice grains and distinguish varieties based on visual characteristics such as color, texture, and shape [6]. By eliminating human bias, computer vision significantly improves the efficiency and accuracy of classification tasks [7,14]. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have further advanced the field, demonstrating an exceptional ability to identify subtle differences in rice varieties [8]. However, earlier studies on CNNs often focus on other agricultural products, such as corn [9], and more tailored approaches are required to address the specific needs of rice classification. Computer vision has found wide applications in agriculture, including

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crop monitoring [10], disease identification [11], and yield estimation [12]. Incorporating rice variety classification into automated systems enables large-scale, real-time assessments, essential for maintaining the integrity and quality of rice products. Additionally, these systems offer a promising solution to issues such as counterfeit rice varieties and fraudulent labeling, which remain prevalent in certain regions [13]. The evolving methodologies in deep learning hold great potential to surpass traditional models in both efficiency and precision, particularly in practical applications. This review provides a comprehensive analysis of the advancements in rice variety classification using computer vision techniques. It focuses on recent developments in image acquisition, pre-processing methods, feature extraction, and classification algorithms that drive the automation of rice variety identification. Additionally, this review addresses key challenges such as the need for larger, more diverse datasets and the impact of environmental variability on classification performance. By synthesizing findings from major studies, the review highlights the strengths and limitations of current methodologies and outlines future directions for enhancing rice classification technologies. This study serves as a valuable resource for researchers and practitioners aiming to advance rice classification systems and contribute to global efforts toward food security and sustainable agriculture.

2. Research questions and motivations

Research questions form the essential foundation for any research study, whether it involves a literature review or an experimental investigation [15]. These questions define the methodological aspects of the research process and guide the exploration of key challenges and opportunities across various disciplines. Below, we outline specific research questions and their corresponding motivations related to rice variety classification, with a particular focus on computer vision techniques. This review compares studies to address these questions and provides insights into the current state of research in this domain.

- i. *How effective are current deep learning models in distinguishing between visually similar rice varieties?*

Motivation: Visually similar rice varieties present a significant challenge for automated classification systems. Investigating the performance of existing deep learning techniques can highlight accuracy gaps and guide the development of improved classification methods.

- i. *What are the most significant image features that contribute to rice variety differentiation?*

Motivation: Understanding the key features (e.g., texture, color, and shape) that drive rice variety classification is crucial for optimizing model architectures. This knowledge can also reduce computational complexity by focusing on the most relevant characteristics.

- i. *Can ensemble models outperform individual models in achieving high accuracy for rice variety classification?*

Motivation: Exploring the potential of combining multiple models into an ensemble could lead to more robust and accurate solutions. While agricultural applications often deal with diverse datasets, the feasibility of applying ensemble techniques in resource-constrained environments remains an important area of investigation.

- i. *How does data augmentation influence the performance of rice variety classification models?*

Motivation: The limited availability of large-scale annotated datasets for rice varieties necessitates data augmentation. Evaluating the impact of these techniques on model performance can provide strategies

to improve accuracy when working with small datasets.

- i. *What are the key challenges in developing scalable and efficient rice variety identification systems for real-world agricultural applications?*

Motivation: Identifying barriers such as limited computational resources, model generalization issues, and environmental variability is essential for advancing practical, scalable systems. Addressing these challenges can make automated rice variety identification more viable for widespread agricultural use.

- i. *How do different imaging conditions (e.g., lighting, camera quality, angles) affect the accuracy of rice variety classification?*

Motivation: Variations in image capture conditions, common in real-world agricultural settings, can significantly influence classification accuracy. Investigating these factors will help create more resilient models capable of functioning under uncontrolled conditions.

- i. *Can transfer learning improve the performance of rice variety identification models with limited training data?*

Motivation: Transfer learning, which involves fine-tuning models pre-trained on large datasets for specific tasks like rice variety identification, offers a promising approach to enhance performance in scenarios with scarce training data.

Moreover, this review focuses primarily on studies utilizing RGB imaging data for rice variety classification. While RGB imaging is one of the most accessible and widely used techniques in this domain, the potential of integrating other imaging methods, such as hyperspectral or multispectral imaging, will also be briefly discussed to provide a broader perspective. Each research question addresses a distinct challenge or opportunity, aiming to advance the development of more accurate and scalable rice classification systems.

3. Methodology for article selection

The present review provides a systematic evaluation of academic research that applies computer vision and machine learning approaches to the automatic classification of rice varieties over the past thirty years. To ensure an exhaustive and transparent evaluation, a systematic identification, selection, and review process was conducted in adherence to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [16]. The PRISMA framework offers a structured and reproducible approach to identifying and appraising relevant research articles, ensuring the inclusion of high-quality and applicable studies. As this topic encompasses diverse areas, including computer vision, artificial intelligence, and agricultural applications, a comprehensive search was conducted in eight major academic databases: Scopus, Web of Science, Science Direct, IEEE Xplore, Springer, Wiley Online Library, arXiv, and ACM Digital Library. These databases were selected for their extensive scope in topics related to computer vision, artificial intelligence, and their agricultural applications, along with their substantial corpus of literature on machine learning and deep learning. We employed advanced search query strings designed to retrieve papers specifically focused on rice variety classification. The following query strings were used:

- i. *Google Scholar*

(intitle:"Rice Variety" OR intitle:"Rice Varieties" OR intitle:"Rice Type" OR intitle:"Rice Types" OR intitle:"Rice" AND (intitle:"Classification" OR intitle:"Deep Learning" OR intitle:"Computer Vision" OR intitle:"Machine Learning" OR intitle:"Identification" OR intitle:"CNN") -Disease -diseases -seed -seeds -quality -leaf -leaves -genetics -genome -paddy -fields -review -yield -nutritional -germ -swarm

-case study -Nutrition -Deficiency after:1994 before:2025)

i. SCOPUS

TITLE=ABS ("Rice" AND ("Variety" OR "Varieties" OR "Type" OR "Types" OR "grains" OR "grain") AND ("Classification" OR "Identification" OR "Recognition" OR "Classify" OR "Identify") AND NOT "Disease" AND NOT "Diseases" AND NOT "Leaf" AND NOT "Leaves" AND NOT "Genetic" AND NOT "Genetics" AND NOT "Seed" AND NOT "Seeds" AND NOT "Paddy" AND NOT "Field" AND NOT "genome" AND NOT "Fields" AND NOT "quality" AND NOT "Food" AND NOT "metabolom") AND PUBYEAR > 1994 AND PUBYEAR < 2025 AND (LIMIT-TO (SUBJAREA, "AGRI") OR LIMIT-TO (SUBJAREA, "COMP")) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "re")) AND (LIMIT-TO (EXACTKEYWORD, "Rice") OR LIMIT-TO (EXACTKEYWORD, "Image Processing") OR LIMIT-TO (EXACTKEYWORD, "Machine Learning") OR LIMIT-TO (EXACTKEYWORD, "Deep Learning") OR LIMIT-TO (EXACTKEYWORD, "Convolutional Neural Network") OR LIMIT-TO (EXACTKEYWORD, "Rice (Oryza Sativa L.)") OR LIMIT-TO (EXACTKEYWORD, "Grain") OR LIMIT-TO (EXACTKEYWORD, "Image Classification") OR LIMIT-TO (EXACTKEYWORD, "Image Segmentation") OR LIMIT-TO (EXACTKEYWORD, "Computer Vision") OR LIMIT-TO (EXACTKEYWORD, "Rice Grains") OR LIMIT-TO (EXACTKEYWORD, "Convolutional Neural Networks") OR LIMIT-TO (EXACTKEYWORD, "Classification") OR LIMIT-TO (EXACTKEYWORD, "Feature Extraction") OR LIMIT-TO (EXACTKEYWORD, "Support Vector Machines")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (PUBSTAGE, "final"))

i. Web of Science

TS=("Rice" AND ("Variety" OR "Varieties" OR "Type" OR "Types" OR "Grains" OR "Grain") AND ("Classification" OR "Identification" OR "Recognition" OR "Classify" OR "Identify") NOT ("Disease" OR "Diseases" OR "Leaf" OR "Leaves" OR "Genetic" OR "Genetics" OR "Seed" OR "Seeds" OR "Paddy" OR "Field" OR "Genome" OR "Fields" OR "Quality" OR "Food" OR "Metabolom")) AND PY=(1995–2024) AND WC=("Agronomy" OR "Computer Science Artificial Intelligence") AND (DT=("Article" OR "Review")) AND LA=("English")

To conduct an advanced search for relevant research articles, specific query strings were used across multiple academic databases, incorporating carefully chosen keywords and filters tailored to the study's focus. The keywords primarily centered around terms like "Rice Variety," "Rice Varieties," "Rice Type," "Rice Types," and related phrases, ensuring coverage of a broad spectrum of studies addressing rice classification tasks. Additional focus terms included "Classification," "Identification," "Recognition," "Deep Learning," "Machine Learning," "Computer Vision," and "Convolutional Neural Networks (CNNs)," targeting works leveraging advanced computational techniques for rice grain analysis. Exclusion criteria were applied to omit unrelated topics such as "Disease," "Seed," "Leaf," "Genetics," "Paddy," "Fields," "Quality," and "Nutritional Aspects," ensuring a streamlined focus on rice classification using machine learning and computer vision. Temporal filters limited the search to studies published between 1995 and 2024, reflecting the timeframe of interest. The searches were conducted in English-language articles to maintain consistency and accessibility. Subject-specific and document-type filters were also utilized. In SCOPUS, for example, searches were restricted to subject areas like "Agricultural and Biological Sciences" and "Computer Science," and document types were limited to research articles and reviews. Similarly, in Web of Science, the scope was narrowed to fields such as "Agronomy" and "Artificial Intelligence," focusing on articles and reviews. Exact keywords, including "Image Processing," "Feature Extraction," "Image Classification," and "Support Vector Machines," further ensured alignment with the study's goals.

A total of 3037 records were retrieved from the academic databases, supplemented by 2208 additional records identified through web searches and citation tracking of key studies. After the removal of 1147

duplicate entries, 4098 unique records remained. Automatic filtering based on predefined exclusion criteria, such as non-English papers, studies lacking full-text access, and articles outside the scope of rice variety classification, resulted in the elimination of 1816 records. To ensure relevance and quality, strict inclusion and exclusion criteria were applied:

- Studies must have been published between 1995 and 2024, as, to the best of our knowledge, no published research on rice variety classification exists before 1995.*
- The focus must specifically be on rice variety classification using computer vision, machine learning, or deep learning techniques.*
- Studies related to rice diseases, seeds, leaves, or genetics were excluded.*
- Articles must be written in English and provide full-text access.*

Studies that failed to meet these criteria or focused on non-agricultural applications were excluded. After applying these criteria, 35 records were shortlisted. However, two studies were later excluded due to irrelevance, leaving 33 full-text articles for detailed review. Of these, 29 studies were ultimately included in the final review based on their adherence to the selection criteria. The relatively low number of studies included in the final review, despite the large number of initial records (5245), can be attributed to the highly specific focus of this review. Only studies that directly addressed rice grain variety classification based on the visual features of the grains were considered. Research focusing on other aspects, such as rice diseases, seeds, leaves, genetics, or applications beyond grain classification, was excluded. This strict focus ensured that the review concentrated exclusively on studies directly relevant to the intended research objective, providing a more targeted and meaningful analysis.

During the review process, data extraction focused on pre-processing procedures, feature extraction techniques, classification algorithms, and image capture methodologies. Particular attention was given to studies documenting significant advancements in deep learning approaches, such as Convolutional Neural Networks (CNNs), as well as hybrid techniques combining machine learning and ensemble methods. Each study was evaluated in terms of accuracy, efficiency, and scalability for real-world agricultural applications. Table 1 presents meta-information about the selected research articles. Notably, only 7 of the selected studies included a result comparison with prior research, and 6 studies were supported by proper funding. Fig. 1 illustrates the distribution of studies across journals and conferences, with Elsevier leading in journal publications and IEEE dominating conference contributions. Scopus and Web of Science indexing are consistent, highlighting the prominence of these platforms in research dissemination.

The PRISMA 2020 guidelines were strictly followed throughout the review process, providing a structured approach to identifying, screening, and assessing relevant studies while ensuring transparency and reproducibility. Fig. 2 (Graphical Abstract) and Fig. 3 (PRISMA 2020 flowchart) visually summarize the systematic process undertaken during this review.

4. Rice varieties classification approaches

4.1. Dataset imaging approaches

The image capture methods used in these studies show a variety of strategies tailored to the specific goals and difficulty levels associated with rice grain classification. The use of RGB, hyperspectral, and gray-scale images, along with different pre-processing methods and considerations for lighting and resolution, emphasize the importance of adapting image capture techniques to suit the particular requirements of the classification task. Below is a detailed discussion of the techniques employed, along with their advantages and limitations in Table 2.

Table 1
Meta Information for Final Selected Articles for Review.

Year	Title	Journal Name	Citation (Till 2024)	Scopus	Web of science
1996	Two-Dimensional Image Analysis of The Shape of Rice and Its Application to Separating Varieties [17]	Journal of Food Engineering	161	Yes	Yes
2002	Classification of Rice Grains Using Fuzzy Art Map Neural Network [18]	Asia-Pacific Conference on Circuits and Systems	10	Yes	
2004	On The Use of Counter Propagation Artificial Neural Networks to Characterize Italian Rice Varieties [19]	Analytica Chimica Acta	50	Yes	Yes
2007	characterization and identification of rice grains through digital image analysis [20]	2007 IEEE Instrumentation & Measurement Technology Conference IMTC 2007	50	Yes	
2008	Classification of Philippine Rice Grains Using Machine Vision and Artificial Neural Networks [21]	World Conference on Agricultural Information and IT	83		
2011	Classification of Rice Varieties Using Optimal Color and Texture Features and BP Neural Networks [22]	7th Iranian Conference on Machine Vision and Image Processing	47	Yes	
2012	Application of Imperialist Competitive Algorithm for Feature Selection: A Case Study on Bulk Rice Classification [23]	International Journal of Computer Applications	36		
2012	Design of an Expert System for Rice Kernel Identification Using Optimal Morphological Features and Back Propagation Neural Network [24]	International Journal of Applied Information Systems	35		
2013	Classification And Grading Rice Using Multi-Class SVM [25]	International Journal of Scientific and Research Publications	113		
2014	Classification Of Basmati Rice Grain Variety Using Image Processing and Principal Component Analysis [26]	International Journal of Computer Trends and Technology	46		

Table 1 (continued)

Year	Title	Journal Name	Citation (Till 2024)	Scopus	Web of science
2015	Geometric Feature Extraction of Selected Rice Grains Using Image Processing Techniques [28]	International Journal of Computer Applications	19		
2016	Efficient Technique for Rice Grain Classification Using Back-Propagation Neural Network and Wavelet Decomposition [29]	IET Computer Vision	59	Yes	Yes
2017	A Deep Convolutional Neural Network Architecture for Boosting Image Discrimination Accuracy of Rice Species [30]	Food and Bioprocess Technology	93	Yes	Yes
2017	Classification Of Rice Grain Varieties Arranged in Scattered and Heap Fashion Using Image Processing [31]	Ninth International Conference on Machine Vision (ICMV 2016)	7		
2018	Analysis Of Feature Extraction and Classification of Rice Kernels for Myanmar Rice Using Image Processing Techniques [32]	International Journal of Scientific and Research Publications	24		
2019	Classification Of Rice Varieties Using Artificial Intelligence Methods [33]	International Journal of Intelligent Systems and Applications in Engineering	135	Yes	
2020	Contrastive Analysis of Rice Grain Classification Techniques: Multi-Class Support Vector Machine Vs Artificial Neural Network [34]	IAES International Journal of Artificial Intelligence	42	Yes	
2020	Hyperspectral Imaging for Accurate Determination of Rice Variety Using a Deep Learning Network with Multi-Feature Fusion [35]	Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy	87	Yes	Yes
2021	Classification Of Rice Varieties with Deep Learning Methods [36]	Computers and electronics in agriculture	166	Yes	Yes
2021	Machine Vision Approach for Classification of Rice Varieties	International Journal of Food Properties	28	Yes	Yes

(continued on next page)

Table 1 (continued)

Year	Title	Journal Name	Citation (Till 2024)	Scopus	Web of science
2021	Using Texture Features [27] Determination Of Effective and Specific Physical Features of Rice Varieties by Computer Vision in Exterior Quality Inspection [37]	Selcuk Journal of Agriculture and Food Sciences	17		
2022	Classification Of Rice Using Genetic Fuzzy Cascading System [38]	Lecture Notes in Networks and Systems	1	Yes	
2022	Computer-Assisted Real-Time Rice Variety Learning Using Deep Learning Network [39]	Rice Science	21	Yes	Yes
2022	Rice Grain Classification Using Convolution Neural Network with Small Dataset [40]	2022 Joint 12th International Conference on Soft Computing and Intelligent Systems and 23rd International Symposium on Advanced Intelligent Systems	0	Yes	
2022	Rice Varieties Classification Using Machine Learning Algorithms [41]	Journal of Pharmaceutical Negative Results	17		
2023	On Application of Lightweight Models for Rice Variety Classification and Their Potential in Edge Computing [42]	Foods	8	Yes	Yes
2023	An Efficient Classification of Rice Variety with Quantized Neural Networks [43]	Electronics	4	Yes	Yes
2024	Classification Of Rice Grain Varieties Using Ensemble Learning and Image Analysis Techniques [44]	Indonesian Journal of Data and Science	6		
2024	The Rapid Non-Destructive Differentiation of Different Varieties of Rice by Fluorescence Hyperspectral Technology Combined with Machine Learning [45]	Molecules	3	Yes	Yes

- i. **RGB Images:** Standard RGB (Red, Green, Blue) images, which capture the visible color spectrum, were used in most studies for rice grain classification. These images provide detailed information about the grains' color, texture, and structural patterns. This versatility makes RGB imaging a widely used and cost-effective method in rice variety classification. However, the method is limited by its inability to capture spectral data beyond the visible spectrum, which could be crucial for identifying subtle differences between grain varieties [17–25,34,36,41–44].
 - a. **Grayscale Images:** Grayscale images, often derived from RGB images, were utilized in certain studies to simplify feature extraction and reduce computational complexity. While grayscale images can effectively highlight texture features, they lack the color information necessary for capturing finer distinctions between rice grain varieties. Thus, their usage is generally limited to specific research scenarios [20,21,27,36].
- ii. **Hyperspectral Images:** Hyperspectral imaging (HSI) records spectral data across a wider range, including ultraviolet and infrared wavelengths, providing more comprehensive physical and chemical information about the grains. This allows for more precise classification compared to RGB imaging. However, HSI requires specialized equipment and involves higher computational costs, making it less accessible for practical applications. One study has applied spectral ranges of 400–1000 nm [35] and another study stated their spectral range from 376 to 1000 nm [45] to collect data.
- iii. **Zoom and Focus:** The degree of zoom and focus differed among studies. While some focused on individual rice grains for detailed analysis, others captured bulk samples for segmentation and classification. Close-up imaging of individual grains tends to yield higher accuracy but is more labor-intensive. On the other hand, bulk imaging can enhance efficiency but often introduces challenges related to segmentation and overlapping grains [28,29].
- iv. **Lighting Conditions:** Controlled lighting conditions were frequently emphasized to ensure consistency in image quality and minimize variations that could impact feature extraction and classification accuracy. Uniform and stable lighting is particularly critical for eliminating shadows and reflections, which can obscure important features of the rice grains [26,36,37].
- v. **Image Pre-processing:** Pre-processing techniques, including image segmentation, noise reduction, and background removal, were employed to enhance image quality and simplify feature extraction. While pre-processing improves classification accuracy, it also adds a computational step that may increase processing time in real-time applications [26,37,44].
- vi. **Scattered vs. Heaped Arrangements:** Studies addressing the arrangement of rice grains explored challenges associated with scattered versus heaped formations. Scattered arrangements facilitate easier feature extraction and classification but require careful placement of grains. Heaped arrangements, while more representative of real-world scenarios, often lead to overlapping grains, complicating segmentation and feature extraction [31].
- vii. **Real-time Processing:** A few studies explored real-time classification systems using high-resolution cameras and specialized hardware. These systems aim to improve the efficiency of rice grading and quality control processes. However, implementing real-time processing in practical settings often requires balancing computational speed with classification accuracy [39].

The image dataset acquiring approaches can be generalized into single grain and bulk grain imaging, each serving specific analytical purposes. Single grain imaging [17,20,37,41–44] focuses on capturing individual grains with high clarity and precision, using dimensions such as 640×400 (24-bit) [17], 500×500 (24-bit, cropped) [30], and the widely adopted 250×250 resolution [36,37,41–44]. These images facilitate detailed analysis of grain characteristics, including shape, size,

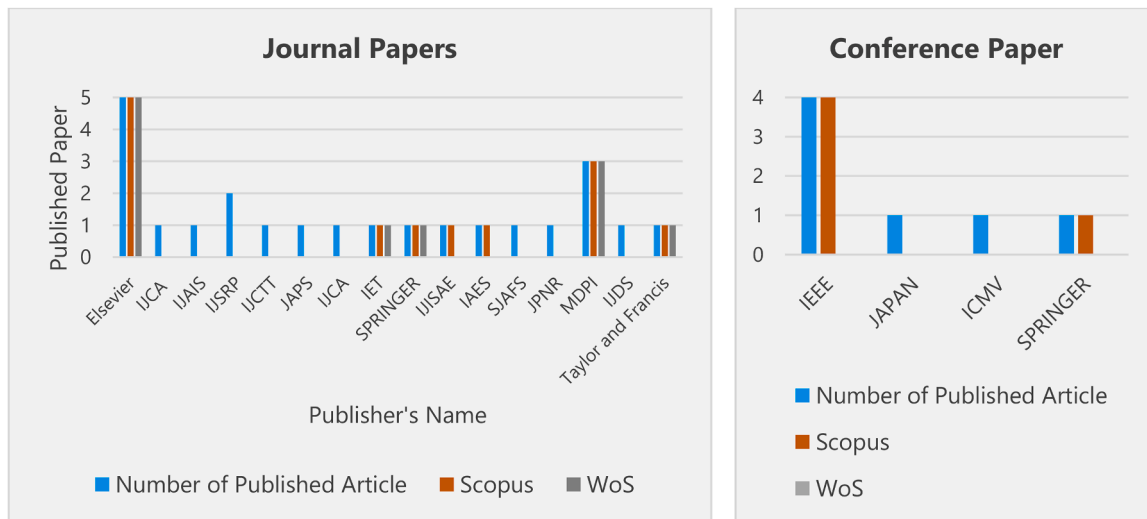


Fig. 1. Distribution of Selected Studies Across Journals and Conferences.

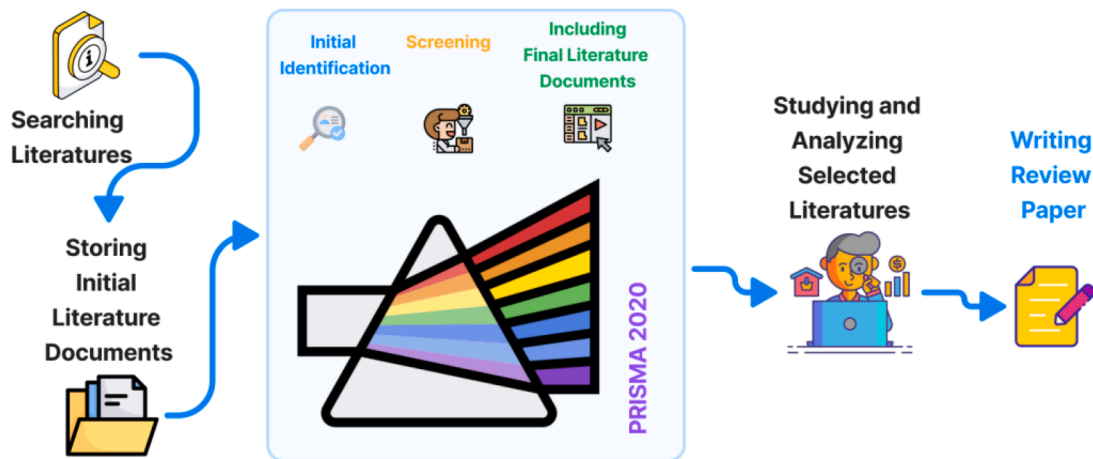


Fig. 2. Graphical Abstract of Review Article Writing Process.

texture, and surface features. Pre-processing methods like binarization [18], boundary detection algorithms [31], and cropping [30] are used to isolate grains and enhance feature extraction. Boundary detection, in particular, effectively delineates grain edges for morphological analysis. In contrast, bulk grain imaging [21,25,27,29,32,16,35,38–40] provides a macro-level view by analysing grain samples collectively. Dimensions for bulk imaging vary from lower resolutions, such as 360×640 (24-bit) [26] and 180×180 [39], to high resolutions like 640×460 , 4000×3000 [29], and 1392×1392 (12-bit) [35]. Preprocessing techniques, including segmentation [22,24–26,32], thresholding, and edge detection [20], are crucial for separating individual grains from the background and each other. Advanced methods, such as MATLAB-based processing [28,33] and partitioning [34], further improve accuracy and efficiency in analyzing grain clusters and distribution patterns.

4.2. Feature extraction

The literature highlighted the use of both traditional morphological traits and advanced deep learning techniques to categorize rice grains. The careful selection and extraction of relevant features—whether manually crafted or learned using Convolutional Neural Networks (CNNs)—is essential for achieving high classification accuracy and for developing reliable automated systems for rice quality evaluation and variety identification.

4.2.1. Morphological and color-texture features

Shape-related features are widely employed in rice grain classification. These include measurements like area, perimeter, major and minor axis lengths, aspect ratio (length & width), compactness, elongation, circularity, rectangularity, and equivalent rectangular dimensions. These features help capture the overall form and proportions of the rice grains, aiding in distinguishing between different varieties [17,19–21, 24,25,28,34].

Size-related features, such as grain length, width, area, perimeter, and Feret diameter, are also commonly used. These features provide quantitative information about the dimensions of the rice grains, which can vary significantly between varieties [20,21,24,27].

Texture features, often derived from the gray-level co-occurrence matrix (GLCM), are utilized to capture the spatial distribution of gray levels in the rice grain images. These features provide insights into the surface properties and patterns of the grains, which can be indicative of variety or quality [22,27,29].

Color features, extracted from various color spaces like RGB (Red, Green, Blue), HSV (Hue, Saturation, Value), CIE $L^*a^*b^*$ (Lightness, Green to magenta, Blue to yellow), $YCbCr$ (Luminance-Chrominance), and XYZ (linear combinations of RGB), are employed to characterize the color variations among rice varieties. These features can be particularly useful in distinguishing between visually distinct rice types [22,23,27, 37].

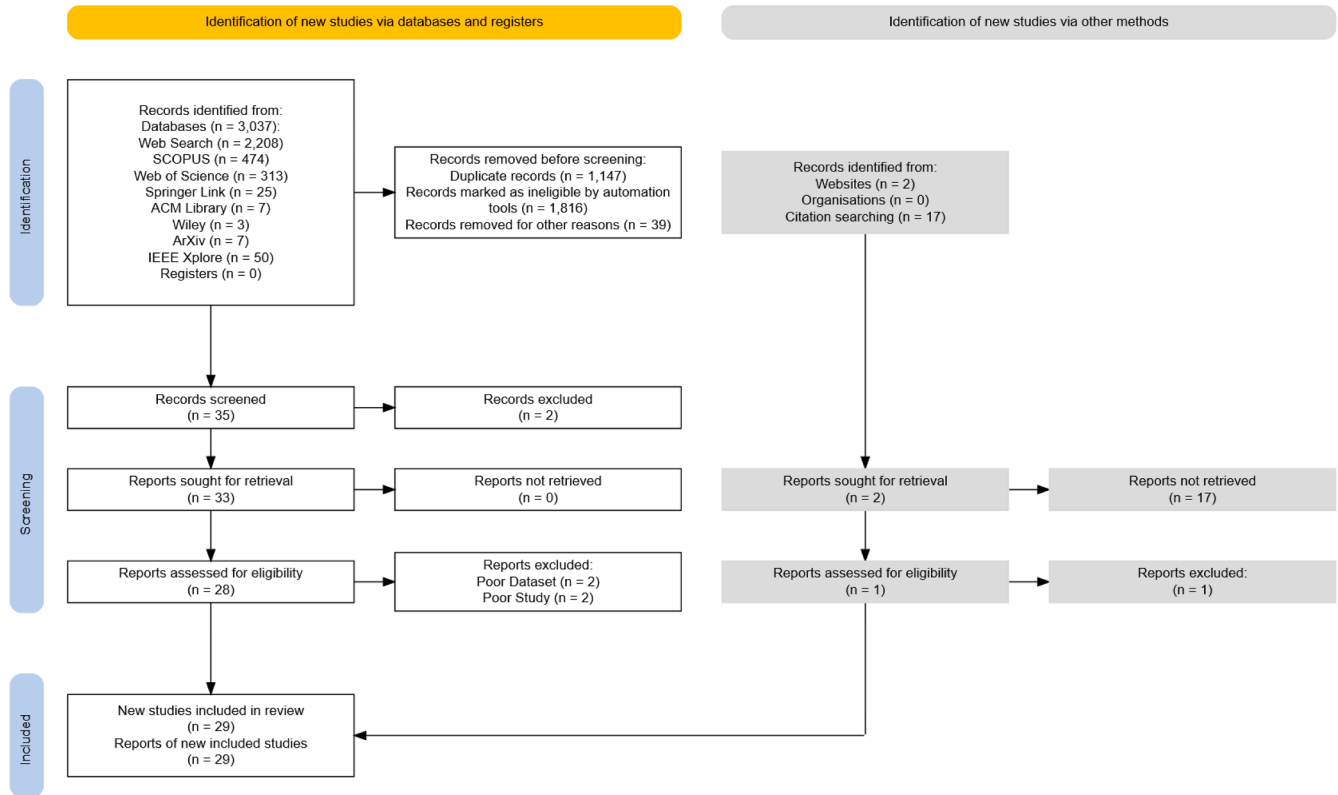


Fig. 3. PRISMA 2020 model for our articles' selection process.

Table 2
Pros and Cons of various Grain Imaging Techniques.

Technique	Pros	Cons
<i>RGB Imaging</i>	- Cost-effective and widely used. - Captures color, texture, and structural patterns.	- Limited to the visible spectrum. - May not distinguish subtle spectral differences effectively.
<i>Hyperspectral Imaging</i>	- Provides comprehensive spectral data beyond the visible range. - Enables precise classification.	- High equipment and computational costs. - Limited applicability in large-scale settings.
<i>Grayscale Imaging</i>	- Simplifies feature extraction. - Reduces computational complexity.	- Lacks color information necessary for comprehensive classification. - Limited standalone utility.
<i>High-Resolution Imaging</i>	- Provides detailed feature data for enhanced accuracy.	- Requires significant computational resources. - Potentially slower processing times.
<i>Bulk Imaging</i>	- Facilitates high-throughput analysis.	- Overlapping grains complicate segmentation and feature extraction.
<i>Zoomed Imaging</i>	- Improves feature extraction accuracy.	- Labor-intensive and less efficient for large-scale applications.
<i>Controlled Lighting</i>	- Ensures consistent image quality. - Reduces variances like shadows and reflections.	- Challenging to maintain in real-world settings.
<i>Real-time Processing</i>	- Enables immediate decision-making for quality control.	- Balances between speed and accuracy remain challenging. - Requires specialized hardware.

4.2.2. Deep features

Several studies applied the Convolutional Neural Network (CNNs) for automatic feature extraction from rice grain images. CNNs are

capable of learning hierarchical representations of images, capturing intricate patterns and textures that may not be easily discernible through manual feature engineering. This approach has shown promising results in improving classification accuracy and reducing the reliance on pre-defined features [30,39,36,40,42].

In some cases [36,40,42], transfer learning was employed, where pre-trained CNN models (e.g., VGG16, MobileNet, NASNet etc.) were utilized as feature extractors. This approach leverages the knowledge learned from large-scale image datasets which enables the effective feature extraction even with limited training data for rice grain classification.

4.3. Classification techniques

The research articles showcased a variety of classification techniques, ranging from traditional machine learning algorithms to advanced deep learning methods. The choice of technique often depended on the specific dataset, features used, and desired trade-off between accuracy and computational efficiency. Deep learning methods, particularly CNNs, generally outperformed traditional approaches, highlighting their potential for revolutionizing rice grain classification and quality assessment in the agricultural industry. Fig. 4 represents all the approaches used in Rice Variety Classification (RVC) in these 29 studies.

4.3.1. Different types of approaches

- Support Vector Machines (SVM):* SVM was a popular choice, utilized in multiple studies [21,19,25,36,34,41,44] for its ability to handle high-dimensional data and complex classification tasks. It was often used in conjunction with feature selection techniques to optimize performance.
- K-Nearest Neighbours (K-NN):* K-NN was employed in some studies [18,34,32] for its simplicity and ease of implementation. It classifies

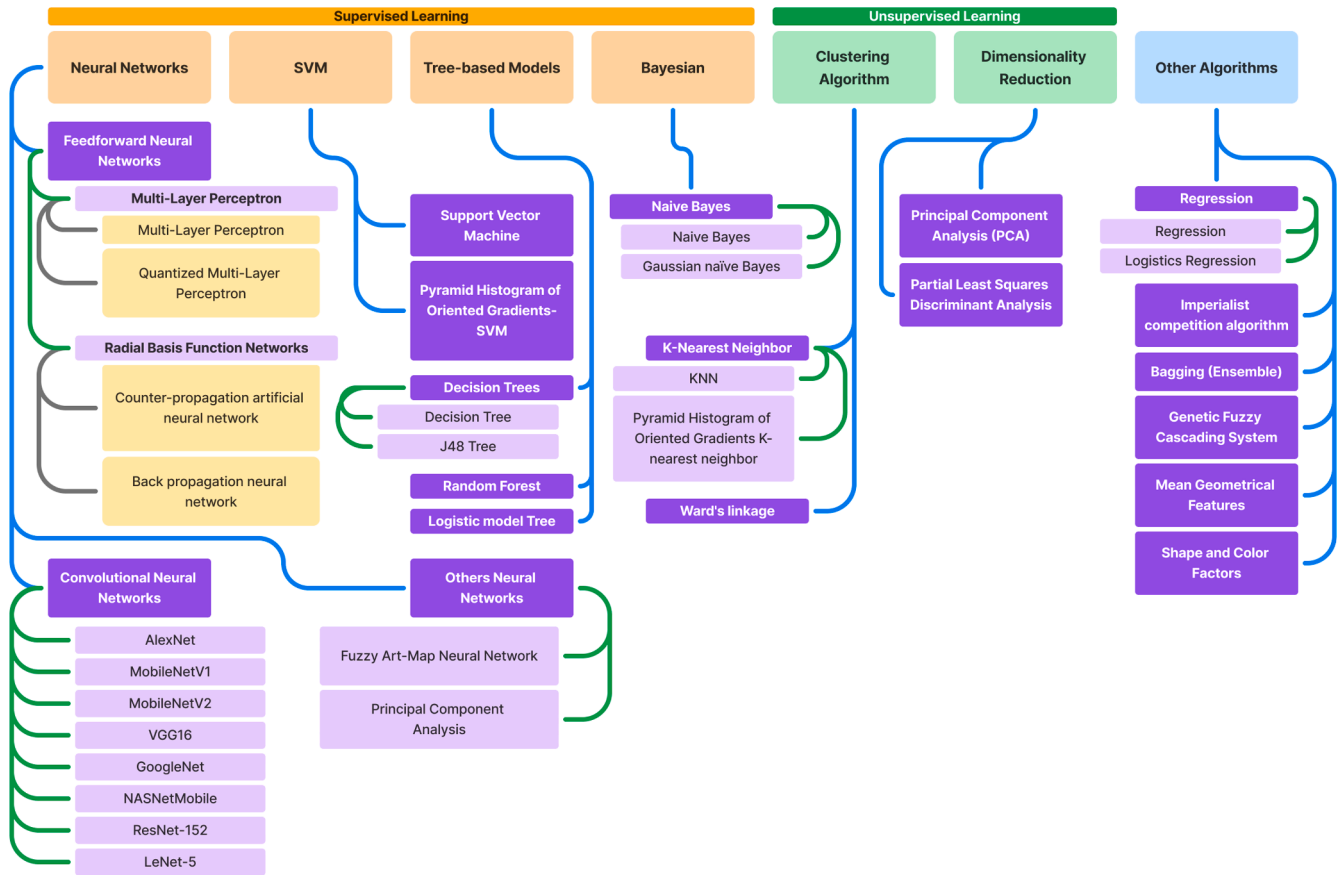


Fig. 4. List of various machine learning approaches used in rice variety classification.

- new data points based on the majority vote of their nearest neighbours in the feature space.
- iii. **Random Forest:** Random Forest, an ensemble learning method, was also utilized for its ability to handle large datasets and reduce overfitting. It constructs multiple decision trees and combines their predictions to improve classification accuracy [41].
 - iv. **Decision Trees:** Decision Trees were used in some studies for their interpretability and ability to handle both categorical and numerical data. They create a tree-like model of decisions and their possible consequences [27,41].
 - v. **Fuzzy ARTMAP Neural Network:** This specialized neural network was employed in one study [18] to address the challenges of real-time rice grain classification. It combines fuzzy logic with adaptive resonance theory to achieve fast learning and high accuracy.
 - vi. **Counter Propagation Artificial Neural Networks (CP-ANN):** CP-ANN, a hybrid neural network architecture, was used in one study [19] to enhance the classification accuracy of rice samples based on physical measurements.
 - vii. **Genetic Fuzzy Cascading System:** This approach combined fuzzy logic with genetic algorithms to improve the explain-ability of the classification model while maintaining high accuracy [38].
 - viii. **Backpropagation Neural Network (BPNN):** BPNN, a classic neural network architecture, was utilized in several studies [24,22,29] for its ability to learn complex patterns and relationships in data. It was often combined with feature extraction techniques like wavelet decomposition to improve classification performance.
 - ix. **Principal Component Analysis (PCA):** PCA was utilized in some studies [26,35,45] for dimensionality reduction, transforming the original feature space into a lower-dimensional space while retaining the most important information. This technique can simplify the classification process and improve computational efficiency.

- x. **Convolutional Neural Networks (CNNs):** CNNs emerged as a powerful tool for rice grain classification, demonstrating high accuracy and the ability to automatically learn features from images. Various CNN architectures, including VGG16, MobileNet, and NASNetMobile, were employed in different studies [30,36,39,40,42].
- xi. **Deep Neural Networks (DNNs):** DNNs, a broader class of neural networks with multiple hidden layers, were also utilized for their ability to learn complex representations from data [39].
- xii. **Ensemble Methods:** Ensemble learning techniques, such as the Bagging meta-estimator, were employed to combine the predictions of multiple base classifiers (e.g., SVM) to improve overall classification accuracy and robustness [27,44].

Table 3. shows a comprehensive pros & cons, suitability and insights from various applied approaches in different studies.

5. Discussion & insights

5.1. Pros and cons of various approaches

All these studies applied different types of unique approaches to extract features, classify variations, speed-up model's performances, and increase accuracies. Table 4 summarizes the overall major approaches and their pros and cons.

5.2. Dataset descriptions and results summary

The datasets analyzed in these studies focus on the classification of various rice varieties through image processing, feature extraction, and machine learning techniques. Experiments were conducted using diverse rice types originating from different regions, with dataset sizes

Table 3
Summary of Machine Learning Approaches for Rice Grain Classification.

Approach	Application Suitability	Technical Insights	Pros	Cons
Support Vector Machines (SVM) [21,19,25,36,34, 41,44]	Best for tasks requiring high-dimensional feature spaces, suitable for quality inspection or classification tasks. May not be ideal for real-time applications.	SVM optimizes the hyperplane to maximize margin between classes. Often combined with feature selection methods to enhance performance.	- Good at handling high-dimensional data. - Effective for complex classification tasks.	- Computationally expensive for large datasets. - Less effective with noisy data or overlapping classes.
K-Nearest Neighbours (K-NN) [18,34,32]	Suitable for small-scale datasets or low-latency tasks where real-time processing is important.	K-NN requires storing the entire training dataset. It's simple but slow as data size grows, making it less scalable for large datasets.	- Simple to implement. - No training phase.	- Slow inference on large datasets. - Sensitive to irrelevant features.
Random Forest [41]	Ideal for large datasets and complex feature spaces where interpretability is needed, such as grading systems.	Constructs a collection of decision trees with random subsets of features, improving robustness. It can handle both regression and classification tasks effectively.	- Reduces overfitting. - Works well on large datasets.	- Model can be complex and hard to interpret.
Decision Trees [27,41]	Suitable for tasks requiring interpretability, like grading and categorization. May not be ideal for highly non-linear data.	The tree splits nodes based on feature value thresholds, providing easy-to-understand classification rules.	- High accuracy. - Easy to interpret and visualize. - Handles both categorical and continuous data.	- Training time can be long for large datasets. - Prone to overfitting. - May not capture complex relationships in data.
Fuzzy ARTMAP Neural Network [18]	Best suited for real-time rice grain classification or applications where speed is crucial.	Fuzzy ARTMAP uses the ART principle to retain memory while learning, making it efficient for dynamic environments.	- Fast learning and high accuracy. - Good for real-time classification tasks.	- Limited usage in large-scale datasets. - Complex to implement.
Counter Propagation Artificial Neural Networks (CP-ANN) [19]	Effective in applications requiring high accuracy based on physical measurements, such as rice grain classification.	CP-ANN uses both unsupervised and supervised phases to enhance learning. Suitable for applications requiring high precision.	- High classification accuracy. - Can work with physical measurements.	- Requires large datasets for optimal performance. - Slower learning phase.
Genetic Fuzzy Cascading System [38]	Suitable for tasks where interpretability and accuracy are crucial, such as grading systems or quality assessment.	This approach uses genetic algorithms to optimize the structure of fuzzy logic systems, providing explainable and robust models.	- Improves model explainability. - Maintains high accuracy.	- Computationally expensive. - Requires fine-tuning of parameters.
Backpropagation Neural Network (BPNN) [24,22, 29]	Best for applications with complex patterns, such as classification based on visual data.	BPNN requires large amounts of data for training and may suffer from overfitting if not properly regulated.	- Capable of learning complex patterns. - Suitable for various types of data.	- Prone to overfitting. - Requires a large dataset for effective training.
Principal Component Analysis (PCA) [26,35, 45]	Ideal for reducing feature space for classification tasks, especially in datasets with high-dimensionality.	PCA transforms the feature space to retain the most important features, simplifying the classification task.	- Reduces dimensionality. - Improves computational efficiency. - Reduces noise.	- Loses some information during dimensionality reduction. - May not be ideal for non-linear data.
Convolutional Neural Networks (CNNs) [30,36, 39,40,42]	Best suited for image-based tasks, such as rice grain classification, quality inspection, or visual recognition.	CNNs work by learning spatial hierarchies of features in images, making them highly effective for visual recognition. They can automatically learn features from raw data.	- High accuracy in image classification. - Automatically learns features. - Good for visual tasks.	- Requires a large amount of labeled data. - Computationally expensive and requires powerful hardware.
Deep Neural Networks (DNNs) [39]	Suitable for complex applications requiring deep feature extraction, such as image-based or high-dimensional classification.	DNNs learn complex non-linear relationships in data, providing high accuracy when trained properly.	- Good for complex data. - High accuracy on large datasets.	- Requires large datasets and high computational resources. - Prone to overfitting without proper regularization.
Ensemble Methods [27,44]	Suitable for applications needing high robustness, such as classification in noisy or imbalanced datasets.	Methods like Bagging and Boosting aggregate the predictions of multiple base classifiers to improve performance and generalization.	- Robust against overfitting. - High accuracy. - Suitable for noisy or imbalanced datasets.	- Can be computationally expensive. - Model interpretation is challenging.

ranging from a few dozen samples to over 75,000. These datasets capture a wide range of characteristics, including physical measurements, color attributes, morphological features, and advanced techniques such as spectral analysis. The rice varieties examined in these studies include *Japonica*, *Indica*, *Javanica*, *Ariete*, *Baldo*, *Balilla*, *Cripto*, *Drago*, *Elio*, *Lido*, *Loto*, *Selenio*, *Thaibonnet*, *Volano*, *Basmati*, *Fajr*, *Hashemi*, *Daneboland*, *Gerde*, *Domsiah*, *Tailandi*, *Mahali*, *Neda*, *Khazar*, *Gharib*, *Ghasrdashti*, *Mohammadi*, *Pusa Basmati 1509*, *Pusa Basmati 1121*, *Pusa 44*, *PR121*, *PR122*, *Glutinous*, *Idly*, *Samba*, *Ponni*, *Paw-San*, *Osmancik*, *Cammeo*, *Brown Rice*, *Simiao*, *Zhefang*, *Ningxia Zhenzhu*, *Yili*, *Xiaozhan*, *Meihe*, *Shanglin*, *Panjin*, *Jingshan Qiaomi*, *Wuchang*, *Arborio*, *Ipsala*, *Jasmine*, *Karacagad*, *Kachi-Kainat*, *Kachi-Toota*, *Kainat-Pakki*, *Super-Basmati-*

Kachi, *Super-Basmati-Pakki*, *Super-Maryam-Kainat*, *Glowing*, *Beras Kam-pung*, *Japan Calrose*, *Beras Perang*, and *Black Wild*. Most studies report classification accuracies exceeding 90 %, underscoring the effectiveness of feature extraction methods such as geometric and texture-based attributes. However, studies leveraging CNN and DNN architectures extract more complex and diverse features automatically, which are not explicitly categorized in traditional feature groups like morphological, color, or texture. To avoid confusion, Fig. 5 focuses solely on manually extracted traditional features, while Table 5 summarizes the dataset descriptions and classification results. Regarding dataset collection, seven distinct methods were identified, including scanning, photography with controlled lighting, hyperspectral imaging, and others,

Table 4
Pros and Cons of Various Approaches.

Categories of Approaches	Imaging Technique	Applied Models and Methods	Extracted Features	Pros	Cons (Limitation)
Image Processing & 2D Shape Analysis [17]	RGB	Dimension, Shape Factors	Area, perimeter, max length, max width, compactness, elongation	- High accuracy (95.45 %) using image processing for objective analysis - non-intrusive and precise measurement	- may overlook depth information and subtle 3D features crucial for certain classifications - Complex rice shapes lead to measurement challenges
Fuzzy ARTMAP Neural Network with Zernike Moments [18]	RGB	Fuzzy Art-map Neural Network	Zernike moments (Morphological)	- Rotationally invariant feature extraction improves classification	- Translation and scale invariance challenges
Counter Propagation Artificial Neural Networks (CP-ANN) [19]	RGB	Counter-propagation artificial neural network	Physical measurements (e.g., size, shape)	- High accuracy (>90 %) - Combines supervised and unsupervised learning for complex data - Fast convergence	- Requires significant computational resources - Limited to physical measurements, may not capture all features
Digital Image Analysis for Rice Identification [20]	RGB, Grayscale	Ward's linkage	Grain length, aspect ratio, compactness, texture	- Cost-effective and non-intrusive method - Introduces novel features like compactness and texture	- Grayscale images limit color-based classification accuracy - Dependent on image quality for robust results
Machine Vision & Artificial Neural Networks (ANNs) [21]	RGB, Grayscale	Multi-Layer Perceptron	13 morphological features	- High classification accuracy (up to 98.76 %) - Automated system reduces human error	- Complexity requires high computational resources - Image quality may impact classification
Color and Texture Feature Extraction & BP-NN [22]	RGB	feed-forward neural network	60 color and texture features	- Combines color and texture features for improved accuracy - Systematic feature selection enhances efficiency	- Image acquisition quality affects robustness - Risk of overfitting with complex neural networks
Imperialist Competitive Algorithm (ICA) for Feature Selection [23]	RGB	Imperialist competition algorithm, Support vector machines	Color, shape, texture attributes	- Systematic feature selection improves classification - Evolutionary optimization methods are effective in agriculture	- Computational resource-intensive method - May miss some relevant non-image-based rice characteristics
Back Propagation Neural Networks (BPNN) with Morphological Features [24]	RGB	feed-forward neural network	18 morphological features	- High accuracy (98.4 %) using morphological features - Reduces computational complexity through feature selection	- Reliance on morphological features limits use of color and texture - Image quality affects robustness
Multi-Class Support Vector Machine (SVM) [25]	RGB	Support Vector Machine	10 geometric features (Morphological, Color and Texture)	- High classification accuracy (86 %) - Robust for practical applications	- Limited ability to capture non-geometric features (e.g., color/texture) - Image quality influences outcomes
Principal Component Analysis (PCA) [26]	RGB	Principal Component Analysis	Morphological Features derived from image processing and PCA	- Effective for dimensionality reduction - Enhances classification efficiency	- Dependent on image quality and lighting uniformity - Possible loss of critical information during dimensionality reduction
Meta-Classifiers [27]	Grayscale	Logistic Model Tree, Regression, J48 Tree, Meta Bagging, Meta Attribute Selected Classifier	43 Morphological, Color and Texture Features	- LMT-T achieved the highest accuracy (97.39 %) - handled noisy data well - J48 Tree and MAS-C are interpretable and lightweight	- computationally expensive for larger datasets - require more computational resources due to their complexity - slightly underperform compared to ensemble-based approaches
Back-Propagation Neural Network (BPNN) & Wavelet Decomposition [29]	RGB	Bayesian, Support Vector Machine, Back Propagation Neural Network, K-Nearest Neighbor	18 color features, 27 texture features, 24 wavelet features	- Robust feature extraction method using wavelet decomposition - High classification accuracy	- Complexity increases computational demands - Focus on specific features may miss other important attributes
Deep Convolutional Neural Networks (CNN) [30,40,36]	RGB, Grayscale	artificial neural networks, Deep Neural Network	Spectral features, morphological features, texture 24 color features, 11 morphological features, 4 shape factors	- High accuracy and robustness against varying image quality - Handles complex classification	- High computational cost and large dataset requirement - Overfitting risk without proper tuning

(continued on next page)

Table 4 (continued)

Categories of Approaches	Imaging Technique	Applied Models and Methods	Extracted Features	Pros	Cons (Limitation)
K-Nearest Neighbours (KNN) Classifier [32]	RGB	K-Nearest Neighbor	Morphological features of Broken Rice	- Non-destructive and efficient method - High accuracy (up to 100 %) for different rice grades	- Relies heavily on specific features like broken rice percentages - Image quality impacts robustness
Fuzzy Cascading System with Genetic Algorithms [38]	RGB	Genetic Fuzzy Cascading System	Morphological	- High accuracy (94 %) - Enhanced interpretability with fuzzy logic	- Complex model design requires significant computational resources
Fluorescence Hyperspectral Imaging & Machine Learning [45,41]	Hyperspectral	logistic regression, support vector machine, Gaussian naïve Bayes, random forest, decision Tree, Partial Least Squares Discriminant Analysis, Back propagation neural network, Random Forest	Morphological and Spectral features across different wavelengths	- Non-destructive and highly accurate classification - Integrates machine learning for spectral feature processing	- Hyperspectral data complexity - Risk of overfitting without large datasets
Ensemble Learning with SVM [44]	RGB	Bagging meta-estimator(Super Vector Machine)	Morphological, Color and Texture	- High classification accuracy (>96 %) - Bagging improves model robustness	- Dataset diversity limitations - May not generalize well to other rice varieties

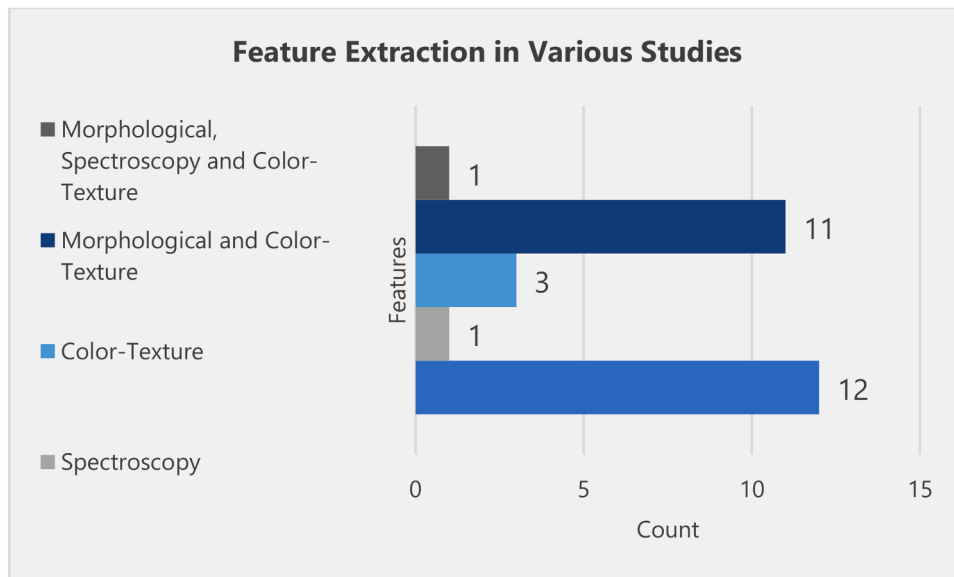


Fig. 5. Combinations of Dataset Feature Extraction in Various Studies.

which are shown in Fig. 6. Fig. 7 visualizes the distribution of datasets by the number of rice varieties, dataset size, and country of origin. For example, one study [21] used 52 rice varieties but had a relatively small dataset size of 5720 images, sourced from the Philippines. Conversely, another study [36] developed a massive dataset of 75,000 images, but it only covered 5 varieties from Turkey. Recently, a comprehensive dataset collection named Aruzz22.5 K was introduced [46,47], which includes 22,500 samples, provides a valuable resource for rice classification research by standardizing the data structure and addressing previous gaps in dataset diversity and size.

5.3. Challenges and gaps

The application of computer vision to classify rice varieties has seen significant progress, as evidenced by the 29 research studies reviewed. However, several challenges and limitations persist, hindering the full potential of this technology in the agricultural sector.

5.3.1. Dataset limitations: a key bottleneck

A consistent challenge across multiple studies is the reliance on limited and often homogeneous datasets. While these datasets are

sufficient for initial model development, they often lack the diversity needed to represent the vast array of rice varieties and real-world conditions. This limitation restricts the generalization capabilities of models, making them less effective when deployed in practical scenarios. For example, models trained on rice varieties from a specific geographical region may struggle to classify rice from other regions due to environmental factors influencing grain characteristics [21]. Additionally, image quality and variability remain significant obstacles. Inconsistent lighting, camera angles, and grain orientations introduce noise and inconsistencies, complicating feature extraction and classification tasks. While some studies employ pre-processing techniques and data augmentation to address these issues, there is a pressing need for robust algorithms capable of handling real-world image variability. For instance, one study [29] highlights the importance of addressing image quality variations to achieve reliable classification outcomes.

5.3.2. Feature extraction and selection

Identifying the most discriminative features for rice variety classification remains a complex challenge. Researchers have explored morphological, color, and texture features, yet determining the optimal combination that balances accuracy and efficiency is an open question.

Table 5
Dataset Description Summary and Extracted Features of various studies.

Dataset Short Description	Dataset Size	Varieties	Extracted Features
Measurements of different rice varieties categorized under Japonica, Indica, and Javanica types [17]	400 samples	4	Area, perimeter, max length, max width, compactness, elongation
Images of various rice grains focusing on rotationally invariant features [18]	Not specified	2	Zernike moments
Italian rice characterized by physical measurements [19]	1779 samples	11	Physical measurements (e.g., size, shape)
Images of eight common rice varieties from UK [20]	~400 samples	8	Grain length, aspect ratio, compactness, texture
Images of 52 rice grain samples from various agroecological zones [21]	52 samples	5	13 morphological features
Images of five rice varieties characterized by color and texture [22]	500 grains	5	60 color and texture features
Bulk rice samples (Fajr, Hashemi, Daneboland, Gerde, Basmati, Domsiah and Tailandi) with various morphological features [23]	2100	7	Color, shape, texture attributes
Images of various Iranian rice kernels characterized by morphological features [24]	1500 kernels	5	18 morphological features
Indian Basmati Rice kernel samples analysed for quality grading [25]	Not specified	1	10 geometric features
Images of Basmati rice varieties captured under controlled conditions [26]	477 kernels	3	Features derived from image processing and PCA
8-bit Gray-scale image of Rice grains featured by Binary, Histogram and Texture [27]	10,800 Kernels	6	43 features
Images of bulk rice grains characterized by color, texture, and wavelet features [29]	4 varieties	4	18 color features, 27 texture features, 24 wavelet features
Diverse images of Japanese rice grain species [30]	5554 samples	3	Spectral features, morphological features, texture
Images of Paw-San rice kernels classified based on quality grades [32]	Not specified	3	Broken rice percentage
Grayscale images of different rice types [38]	3810 samples	2	7 input and 2 output variables in Fuzzy Inference System
Fluorescence hyperspectral data from multiple rice varieties [45]	550 samples	5	Spectral features across different wavelengths
45,000 images of Arborio, Basmati, and Jasmine rice varieties [44]	45,000 samples	3	Image features processed with Bagging and SVM
Hyperspectral images of China rice varieties [35]	4320	10	Spectral reflectance, color, morphology, texture
Images of multiple Malaysian rice varieties categorized under controlled conditions [34]	Not specified	3	Morphological and color features
75,000 images of five Turkish rice varieties [36]	75,000 samples	5	24 color features, 11 morphological features, 4 shape factors
75,000 Turkish rice grain images capturing physical characteristics [37]	75,000 samples	5	106 features (12 morphological, 4 shape, 90 color)

Table 5 (continued)

Dataset Short Description	Dataset Size	Varieties	Extracted Features
Grayscale images of Vietnamese rice grains [39]	575 & 750	5 & 4	Various attributes extracted for classification
Images of Indian rice (Basmathi, Idly, Samba, Ponni and Ponni) grains in scattered and heap arrangements [31]	Not specified	5	Geometric features estimated from images
75,000 samples images of Turkish rice varieties [29]	75,000 samples	5	Features related to rice variety classification
75,000 samples images of Turkish rice varieties [42]	75,000 samples	5	Various extracted features from images
Turkish two types of rice grains to obtain morphological features [33]	3810 images	2	Extracted 7 morphological features
75,000 rice grain images collected from grocery stores [41]	75,000 samples	5	Various attributes for quality control
5 types of Indian Basmati rice image taken for MATLAB processing [28]	2917 grains	5	7 geometry features are extracted
Malaysian rice images capturing various varieties [40]	700	7	Various image features

The ideal feature set likely varies depending on specific rice varieties, imaging conditions, and classification objectives. Future research should prioritize developing systematic approaches for feature selection and engineering to ensure robust performance across diverse scenarios [23]. Deep learning methods, particularly Convolutional Neural Networks (CNNs), have revolutionized feature extraction by automating the process. However, challenges remain in creating models that are efficient and interpretable. The computational demands of deep learning techniques can limit their deployment in resource-constrained environments, while their "black-box" nature hinders interpretability. For critical applications where transparency is necessary, balancing accuracy, efficiency, and interpretability is vital for broader adoption.

5.3.3. Classification algorithms

The choice of classification algorithm plays a pivotal role in the success of rice variety identification systems. Traditional machine learning methods, such as Support Vector Machines (SVM) and Random Forests, are favoured for their interpretability and lower computational demands, making them suitable for smaller datasets or resource-limited environments [25,41]. However, these methods may struggle to capture complex patterns in data compared to deep learning approaches. Deep learning models, particularly CNNs, consistently achieve superior accuracy in rice variety classification by learning intricate features from raw image data. Despite their advantages, high computational requirements and lack of interpretability pose significant challenges for real-world deployment. The exploration of lightweight models and quantization techniques, such as those detailed in a study [42], provides a promising avenue for addressing these issues. Hybrid approaches that integrate the strengths of multiple algorithms, such as combining fuzzy logic with machine learning, demonstrate potential for improving accuracy, efficiency, and interpretability. For instance, a study [38] showcases the effectiveness of hybrid techniques in achieving high accuracy while maintaining model explainability.

5.3.4. Real-World applicability

Bridging the gap between controlled laboratory research and real-world applications remains a substantial challenge. Most studies focus on controlled environments, where variables like image quality and grain arrangements are carefully managed. In real-world scenarios, factors such as lighting variations, background clutter, and overlapping

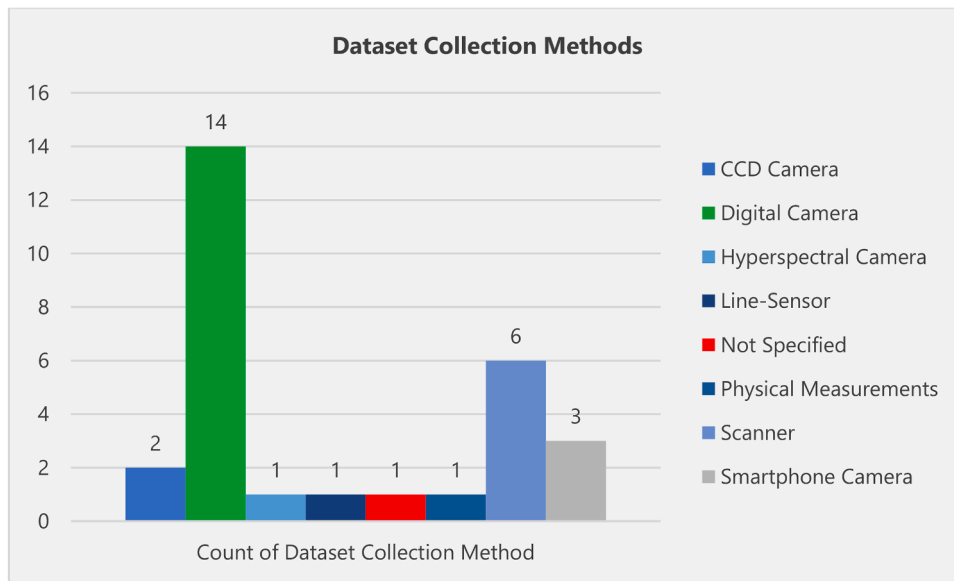


Fig. 6. Dataset Collection Methods.

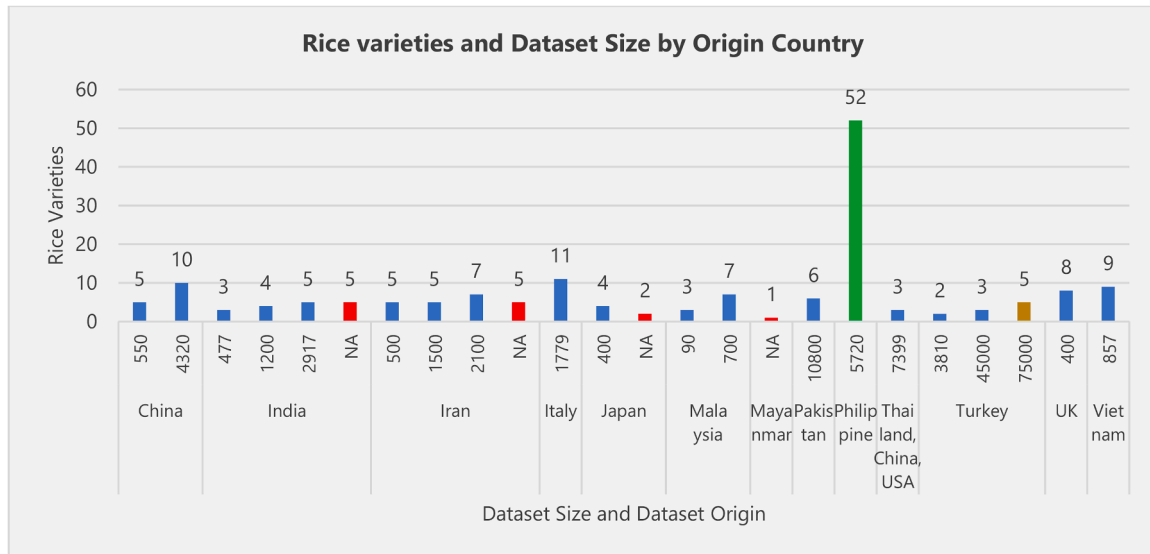


Fig. 7. Summary of rice dataset varieties, size and country origins.

grains create additional complexities. Bhat et al. [31] addresses this issue by proposing techniques to handle overlapping grains, but further research is necessary to enhance robustness under diverse conditions. The successful deployment of computer vision-based systems also requires seamless integration with existing agricultural workflows and infrastructure. This involves developing user-friendly interfaces, compatibility with existing hardware and software, and scalability to handle large datasets. Jeyaraj et al. [38] emphasizes the importance of real-time processing capabilities for practical implementation.

5.4. Results and insights from datasets

Deep learning methods, particularly Convolutional Neural Networks (CNNs), consistently outperformed traditional machine learning algorithms in classification accuracy. Studies utilizing CNNs reported accuracies exceeding 95 %, with some achieving up to 100 % on specific datasets, whereas traditional algorithms generally yielded accuracies between 70 % and 95 % [30]. Lightweight models and quantization

techniques have shown significant potential in reducing computational and memory requirements while maintaining competitive accuracy, making these approaches highly relevant for real-time applications and deployment on resource-constrained devices [43]. Advanced feature selection techniques, such as the Imperialist Competitive Algorithm (ICA), have improved classification performance by identifying the most relevant features and reducing dimensionality, emphasizing the importance of careful feature engineering in optimizing model outcomes [22]. Dataset sizes varied significantly across studies, ranging from tens of thousands of images to just a few hundred samples, with larger datasets enabling more robust and generalizable models. This highlights the critical need for comprehensive dataset collection efforts to enhance model performance and scalability. Table 6 summarizes the applied classification approaches and achieved results in different format for different types and size of rice dataset.

Table 6

Summary of Classification methods and results for different dataset size and varieties.

Year	Data Size	Varieties	Classification Methods	Result Types	Results
1996 [17]	400	4	Dimension and Shape Factor	Probability	68.3 % - 95.4 %
2002 [18]	NA	2	Fuzzy Art-map Neural Network	Accuracy	95.40 %
2004 [19]	1779	11	Counter-Propagation Artificial Neural Network (CP-ANN)	Accuracy	91 %–99 %
2007 [20]	400	8	Unsupervised Clustering Techniques	Dendrogram	Not Specified
2008 [21]	5720	52	Multi-Layer Perceptron	Accuracy	66.67 %–100 %
2011 [22]	500	5	Segmentation And Back Propagation Neural Network-Based Classifier	Accuracy	96.67 %
2012 [23]	2100	7	Imperialist competition algorithm and SVM	F-Measure	90.02 %–95.14 %
2012 [24]	1500	5	Segmentation and Back propagation neural network-based classifier	Accuracy	98.40 %
2013 [25]	Not Specified	1	Multi-Class SVM	Accuracy	>86 %
2014 [26]	477	3	Principal Component Analysis	Accuracy	75 %–80 %
2021 [27]	10,800	6	Logistic Model Tree, Regression, J48 Tree, Meta Bagging, Meta Attribute Selected Classifier	Accuracy	97.4 %
2015 [28]	2917	5	Average of Major axis length	Error Rate	<1.40 %
2016 [29]	1200	4	BPNN, SVM, KNN, Naive Bayes	Accuracy	96.25 %–100 %
2017 [30]	7399	3	DCNN	Accuracy	95.50 %
2017 [31]	Not Specified	5	Scattered and Heap Fashion Image Processing	Scatter Plot	Various Results
2018 [32]	Not Specified	1	KNN	Accuracy	83 %–100 %
2019 [33]	3810	2	LR, MLP, SVM, DT, RF, NB and k-NN	Accuracy	93.02 %
2020 [34]	90	3	SVM and ANN	Accuracy	93.34 %
2020 [35]	4320	10	principal component analysis network	Accuracy	98.57 %
2021 [36]	75,000	5	ANN, DNN, CNN	Accuracy	99.87 %–100 %
2021 [37]	75,000	5	Not Specified	Not Specified	Not Specified
2022 [38]	3810	2	Genetic Fuzzy Cascading System	Accuracy	94.36 %
2022 [39]	857	9	CNN	Accuracy	96.40 %
2022 [40]	700	7	ANN and CNN	Accuracy	88.3 %–99.5 %
2022 [41]	75,000	5	Random Forest	Accuracy	99.85 %
2023 [42]	75,000	5	CNN	Accuracy	99.20 %
2023 [43]	75,000	5	Quantized Neural Network	Accuracy	99.87 %
2024 [44]	45,000	3	Ensemble Learning	Accuracy	96.74 %
2024 [45]	550	5	Principal Component Analysis, BP Neural Network, Random Forest	Accuracy	95.30 %

5.5. State-of-the-Art models and techniques

- i. **Convolutional Neural Networks (CNNs):** CNNs have consistently shown high accuracy in rice variety classification due to their ability to automatically learn intricate features from images. Lin et al. [30] achieved 95.5 % accuracy, outperforming traditional machine-learning approaches. CNNs excel in handling complex patterns such as texture, shape, and color variations in rice grains, making them one of the most effective models for this task.
- ii. **Hybrid Models (Fuzzy Logic & Machine Learning):** A successful hybrid approach integrates fuzzy logic with machine learning to improve classification accuracy while maintaining interpretability. Wee et al. [18] demonstrated notable success by leveraging rotationally invariant Zernike moments for feature extraction, ensuring that rice grains are classified accurately even in different orientations.
- iii. **Imperialist Competitive Algorithm (ICA) for Feature Selection:** In studies using feature selection techniques, the Imperialist Competitive Algorithm (ICA) was particularly effective in optimizing the feature set. The use of ICA, as seen in a study [22] significantly enhanced classification accuracy by selecting the most relevant features from image data, reducing computational complexity and improving model performance.
- iv. **Ensemble Learning with Bagging Meta-Estimator:** Ensemble learning, particularly using a Bagging meta-estimator with Support Vector Machines (SVM), showed promising results in rice variety classification. The ensemble approach aggregated predictions from multiple classifiers to achieve over 96 % accuracy across different rice varieties, as reported in a study [44] This approach enhances both robustness and precision compared to single classifiers. Also, in another study [27] authors used meta-classifiers to classify 6 varieties of rice and achieved 97.4 % accuracy.
- v. **Back-Propagation Neural Network (BPNN) with Wavelet Decomposition:** A successful combination of BPNN and wavelet decomposition was used to extract texture features, improving classification accuracy in challenging conditions, such as varying image quality. Singh et al. [29] achieved high accuracy by leveraging both spatial and frequency domain features.

- vi. **Quantized Neural Networks (QNN):** For resource-constrained environments, Quantized Neural Networks (QNNs) provided an efficient solution. Tasci et al. [43] achieved over 99 % accuracy while minimizing memory usage and computational power, making QNNs ideal for mobile and IoT applications where computational resources are limited.
- vii. **Deep Learning with Hyperspectral Imaging:** Weng et al. [35] integrated spectral, morphological, and textural data to improve classification accuracy. This novel deep learning architecture achieved higher accuracy compared to traditional methods by using hyperspectral data to capture more detailed and subtle differences between rice varieties.
- viii. **Meta-Learning Approaches for >98 % Accuracy:** Meta-learning approaches, which aim to adapt models to new tasks with minimal retraining, were explored in a study [41]. These models incorporated advanced feature extraction and selection techniques to achieve accuracy rates over 98 %, making them suitable for applications requiring high precision.

6. Future directions

The rapid advancements in computer vision and machine learning have created numerous opportunities for improving rice variety classification systems. To further develop this area of research, the following innovative directions should be explored:

- i. **Hybrid pre-processing and segmentation techniques:** Traditional pre-processing methods, such as image smoothing, binarization, and noise reduction, have been widely used in rice image preparation. However, irregular grain shapes and varying textures demand more advanced approaches. Hybrid techniques that combine traditional methods with novel approaches, such as wavelet decomposition and edge-aware algorithms, can better preserve intricate grain details while reducing segmentation time complexity. AI-driven segmentation methods that intelligently adapt to image inconsistencies, such as variations in lighting or grain orientation, represent a promising avenue for future research.

- ii. *Multi-modal feature extraction and next-generation techniques:* Current feature extraction methods often rely on a narrow set of input types. Future studies should explore multi-modal inputs that integrate data from RGB, hyperspectral, and thermal images to capture a broader range of grain characteristics. Advanced deep learning models can incorporate morphological, color, texture, and chemical properties for a more comprehensive representation. Graph-based feature extraction methods, which capture complex spatial relationships between grain parts, offer additional potential for improving classification accuracy.
- iii. *Ensemble learning and adaptive classifiers:* Ensemble learning methods, such as bagging, boosting, and stacking, remain underutilized in rice classification. Combining diverse models, such as Random Forest, Support Vector Machines (SVMs), and Convolutional Neural Networks (CNNs), into unified adaptive systems could significantly enhance performance. These adaptive classifiers should dynamically adjust to different datasets or rice varieties without requiring manual reconfiguration. Real-time adaptive ensemble systems could improve classification reliability during field deployment by leveraging feedback mechanisms.
- iv. *Explainable AI for trustworthy models:* Explainable AI (XAI) has the potential to improve the transparency and trustworthiness of rice classification systems. Developing interpretable CNN architectures that highlight key regions of the grains influencing classification decisions can help users, such as farmers and quality inspectors, understand model outputs. This will aid in integrating AI solutions into agricultural workflows while ensuring human verification of decisions.
- v. *Development of large-scale, diverse datasets:* The scarcity of high-quality datasets limits progress in rice classification. Future efforts should focus on creating open-source, standardized datasets that include a wide variety of rice types, particularly rare or region-specific varieties. These datasets should account for environmental variability, imaging angles, and natural noise (e.g., dust or mixed grains). Techniques like generative adversarial networks (GANs) can be employed to augment datasets by synthesizing realistic rice images. Collaborative initiatives among academic institutions, agricultural organizations, and governments could accelerate the development and sharing of these datasets.
- vi. *Transfer learning and domain adaptation:* Transfer learning, where models pre-trained on general datasets are fine-tuned for specific tasks, offers a practical solution to address data scarcity in rice classification. Deep transfer learning can reduce training data requirements while achieving high accuracy. Domain adaptation techniques can enhance model robustness, enabling better performance across diverse geographical regions and environmental conditions.
- vii. *Integration of machine learning with IoT for real-time classification:* Integrating Internet of Things (IoT) technologies with machine learning models can facilitate real-time rice classification in the field. Lightweight models deployed on IoT devices such as drones or mobile sensors can capture rice images in real time and provide instant feedback. Using edge computing and cloud-based architectures, these devices can aggregate data, improving overall accuracy and reducing latency. This approach could revolutionize quality control processes in rice production and supply chains.
- viii. *Federated learning for collaborative model development:* Federated learning enables collaboration among multiple data sources without sharing sensitive information. Decentralized training of models on local datasets (e.g., from different farms) allows the creation of a global model that benefits from diverse and distributed data. This approach ensures data privacy and reduces biases associated with isolated datasets while improving model robustness and accuracy.
- ix. *Cross-disciplinary research and ethical AI:* Future research should integrate knowledge from genomics, metabolomics, and computer vision to classify rice varieties based on both visual and non-visual characteristics. Ethical AI practices must also be prioritized, ensuring fair and unbiased classification models that account for

underrepresented rice varieties. Additionally, energy-efficient AI systems should be developed to support sustainability in resource-constrained farming communities.

7. Conclusion

This state-of-the-art review has provided a comprehensive examination of the current state of rice variety classification using computer vision and machine learning techniques. The key findings highlight the growing potential of advanced technologies like Convolutional Neural Networks (CNNs), hybrid feature extraction methods, and ensemble learning in automating and improving the accuracy of rice grain classification. However, several challenges persist, including the lack of large, diverse, and standardized datasets, limited innovation in pre-processing and segmentation techniques, and the need for more interpretable models to gain trust in real-world agricultural applications. We identified several critical gaps in the existing research, such as the reliance on small datasets, which limits the generalizability of models across different geographical regions and environmental conditions. There is also noticeable underuse of deep learning models, despite their proven success in other domains, and a lack of efforts in developing novel pre-processing methods tailored specifically for the complexities of rice grains. Furthermore, while ensemble learning and hybrid approaches have shown promise, there is ample room for exploring more adaptive and scalable models that can handle diverse real-world conditions. To advance this field, future research should focus on developing more robust pre-processing techniques, multi-modal feature extraction approaches, and the integration of novel machine learning architectures like deep transfer learning and hybrid classifiers. The creation of large, open-source rice image datasets that represent a wide range of rice varieties and environmental factors will be crucial for enhancing the accuracy and scalability of these systems. Additionally, explainable AI models will play a key role in making these systems more transparent and user-friendly, facilitating broader adoption across the rice production and quality assurance industries. By addressing these challenges and embracing emerging trends, rice variety classification systems can be transformed into powerful tools that not only improve agricultural productivity but also ensure food security and quality on a global scale.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Md. Masudul Islam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Galib Muhammad Shahriar Himel:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis. **Md. Golam Moazzam:** Supervision, Project administration, Conceptualization. **Mohammad Shorif Uddin:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.atech.2025.100788](https://doi.org/10.1016/j.atech.2025.100788).

Data availability

No data was used for the research described in the article.

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